A $Pt/In_2S_3/CdS/Cu_2ZnSnS_4$ Thin Film as an Efficient and Stable Photocathode for Water Reduction under Sunlight Radiation

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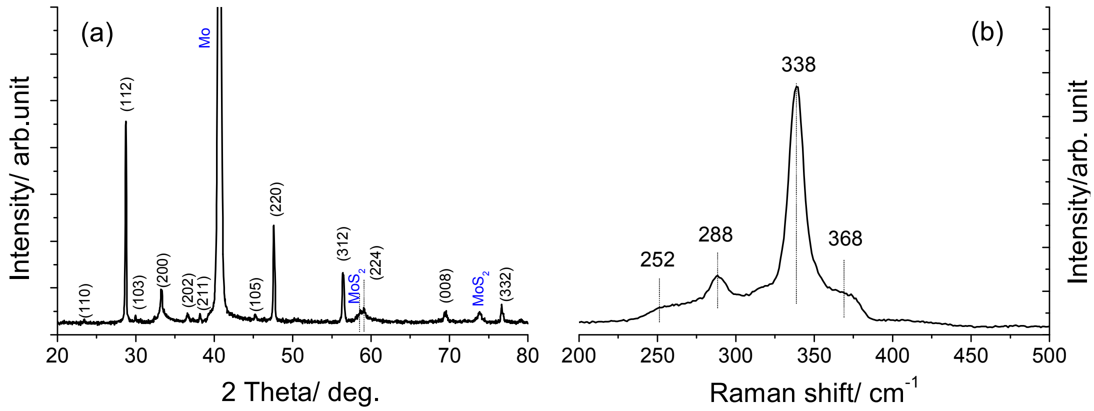

Figure S1: XRD and Raman spectrum of the as prepared CZTS/Mo/glass samples. The XRD pattern of the film matched well with the kesterite structure of CZTS (JCPDS 26-0575): reflections corresponding to (110), (112), (103), (200), (202), (211), (105), (220), (312), (224), (008) and (332) planes were detected without any significant reflections of other compounds except for those derived from the bottom Mo layer. Corresponding Raman spectra showed an intense peak centered at 338 cm^{-1} along with three weak peaks located at 252 cm^{-1}, 288 cm^{-1}, and 368 cm^{-1}; these Raman signals are in agreement with those derived from CZTS. [1-3]

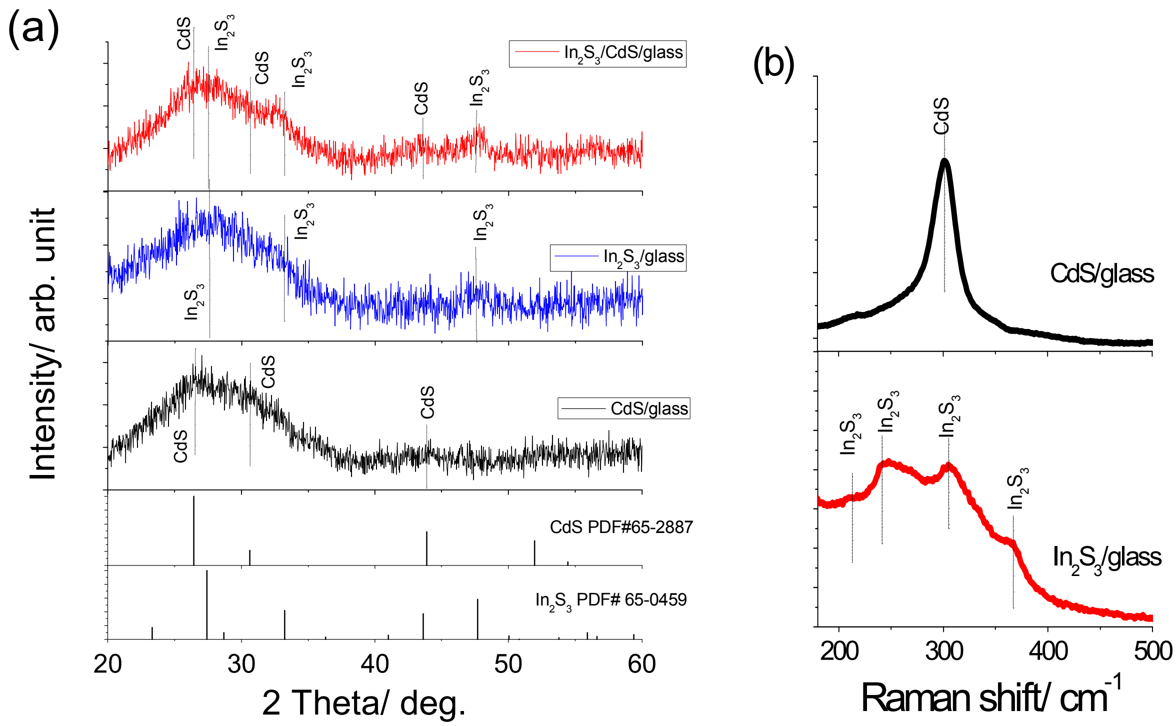

Figure S2: XRD patterns (a) and Raman spectra (b) of CdS/glass and In_2S_3/glass samples. Raman peaks that were observed at 217, 243, 305 and 367 cm^{-1} in Figure S2b (lower) are derived from β-In_2S_3.[4,5] In accordance with literature studies,[6-8] the Raman peak detected from CdS/glass at 305 cm^{-1} shown in Figure S2b (upper) is the LO mode Raman reflection of β-CdS.

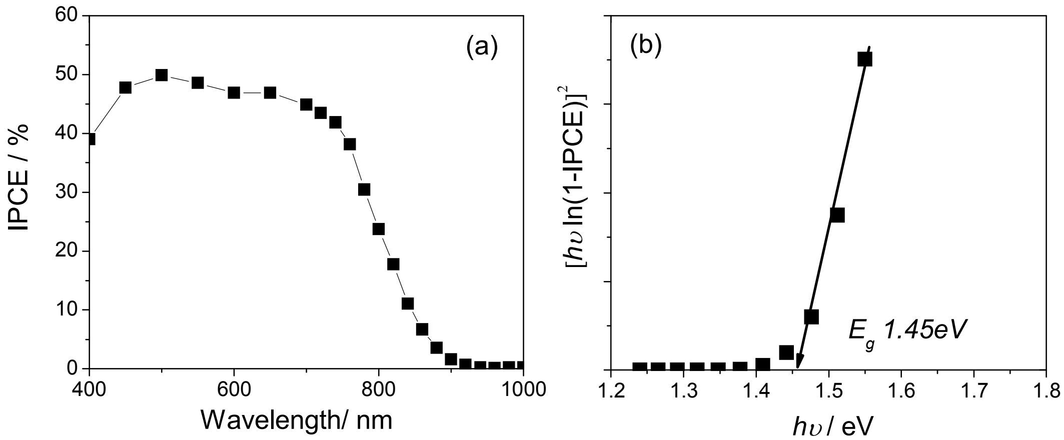

Figure S3: a) IPCE spectraum of $Pt/In_2S_3/CdS/CZTS$ photocathode in a 0.2 mol dm^{-3} Na_2HPO_4/NaH_2PO_4 solution (pH 6.5) at 0 V_{RHE}. b) Plots of $[hv \ln(1-IPCE)]^2$ vs hv for the $Pt/In_2S_3/CdS/CZTS$ film. An intersect line denotes the E_g value.

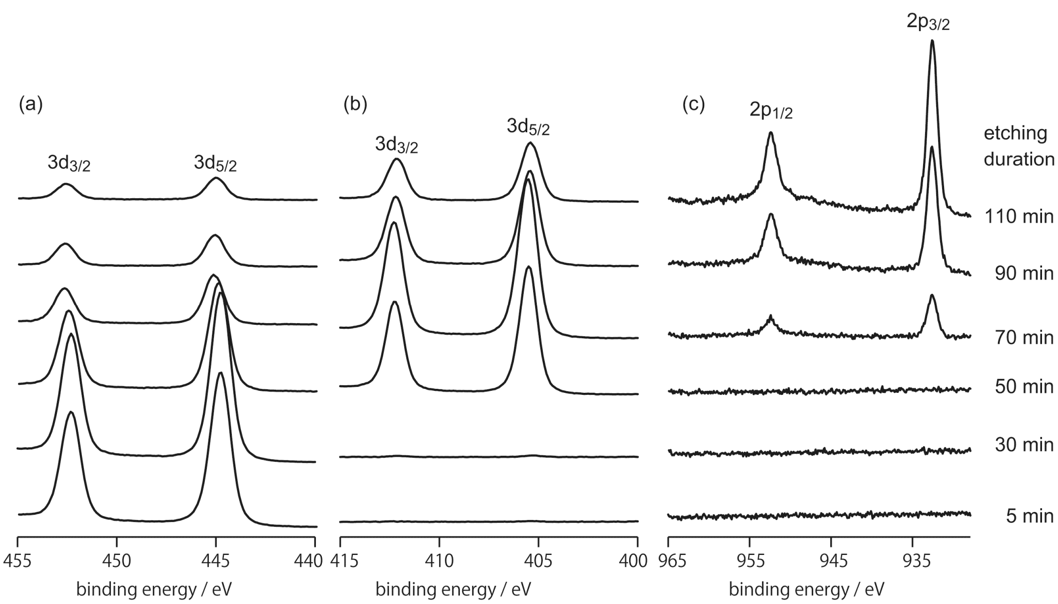

Figure S4: In3d (a), Cd3d (b), and Cu2p (c) XP spectra of $In_2S_3/CdS/CZTS$ heterostructures obtained after Ar^+ ion etching for various durations. When the measurement was performed after 5-min Ar^+-etching, there were intense signals of In without any signals derived from Cd and Cu components, indicating full coverage of the In_2S_3 layer. Intensities of Cd signals gradually increased with extending durations of Ar^+-etching up to 70 min accompanied by appreciable decreases in In signals. Further Ar^+-etching resulted in appreciable reduction of Cd signals as well as appearances of Cu signals derived from the CZTS layer. These results indicate formation of the In_2S_3/CdS double layer on the CZTS film, as expected from SEM results shown in Figures 1c,d and 2c,d.

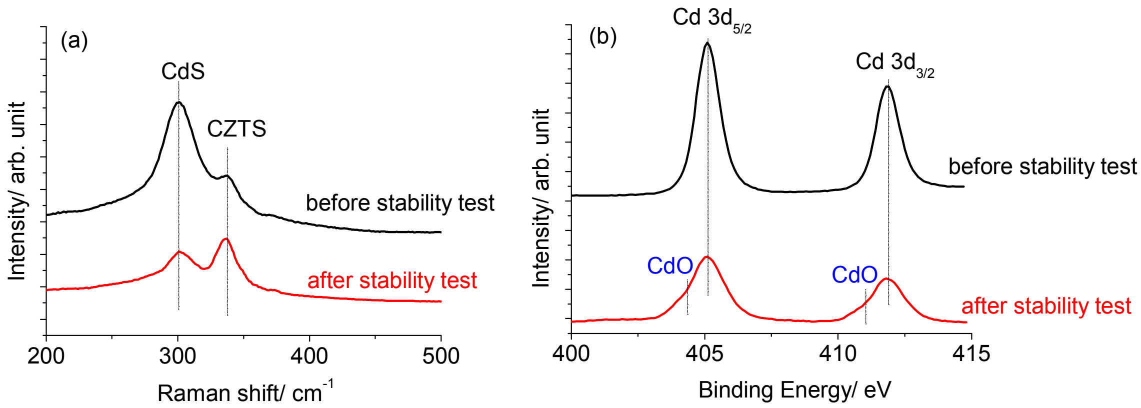

Figure S5: Raman spectra of Pt/CdS/CZTS before and after durability test.

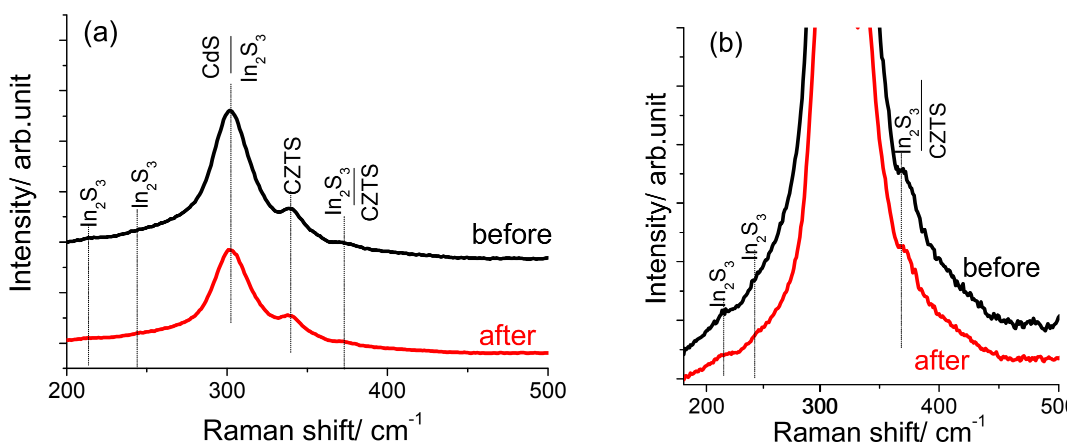

Figure S6: (a) Raman spectra of Pt/In_2S_3/CdS/CZTS before and after durability test, and (b) amplified data of Figure S4a.

References

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